**Project Design Phase-I**
**Proposed Solution Template**

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| Date | 9 March 2026 |
| Team ID | PNT2022TMIDxxxxxx |
| Project Name | Malaria Detection Using Deep Learning |
| Maximum Marks | 2 Marks |

**Proposed Solution Template:**

Project team shall fill the following information in proposed solution template.

| S.No. | Parameter | Description |
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| 1. | Problem Statement (Problem to be solved) | Malaria is a serious infectious disease that affects millions of people worldwide, especially in tropical and developing regions. Early and accurate diagnosis is crucial for effective treatment and reducing mortality. Traditionally, malaria detection is performed through manual microscopic examination of blood smear samples by trained pathologists.<br>However, this method has several limitations such as time consumption, dependency on skilled experts, and the possibility of human error. In many rural or remote areas, trained medical professionals may not be available, which delays diagnosis and treatment.<br>Therefore, there is a need for an automated and reliable system that can assist in detecting malaria infections quickly and accurately. |
| 2. | Idea / Solution description | The proposed solution is a malaria detection system that uses deep learning techniques to analyze microscopic blood smear images and identify malaria-infected cells automatically.<br>The system is trained using a dataset of labeled blood cell images and applies deep learning models such as Custom CNN, MobileNetV2, and EfficientNetB0 to classify the images into two categories: Parasitized (infected) and Uninfected (healthy).<br>A Flask-based web application is developed where users can upload blood cell images. The backend processes the image using the trained deep learning model and provides an instant prediction result. This automated system helps in faster diagnosis and assists healthcare professionals in decision-making. |
| 3. | Novelty / Uniqueness | The uniqueness of the proposed system lies in the integration of deep learning models with a web-based application for automated malaria detection. |

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| | | Key innovative aspects include:<br>• Implementation of multiple deep learning models (Custom CNN, MobileNetV2, EfficientNetB0) for performance comparison.<br>• Automated image analysis that reduces the need for manual microscopic examination.<br>• A user-friendly web interface that allows easy image upload and instant prediction.<br>• A scalable system that can be further integrated with medical imaging devices for automated diagnosis. |
| 4. | Social Impact / Customer Satisfaction | The proposed system can have a significant positive impact on healthcare services.<br>• It can assist doctors and laboratory technicians in diagnosing malaria more efficiently.<br>• It can help provide faster diagnostic support in rural and remote areas where expert pathologists may not be available.<br>• The system reduces diagnosis time, enabling quicker medical treatment for patients.<br>• It can support large-scale malaria screening programs and contribute to improving public health.<br>Overall, the solution improves healthcare accessibility, efficiency, and diagnostic support. |
| 5. | Business Model (Revenue Model) | The proposed system can be implemented as a software-based healthcare diagnostic solution. Possible revenue models include:<br>• Subscription Model: Hospitals and diagnostic laboratories can subscribe to the platform for continuous access to the malaria detection system.<br>• Software Licensing: Healthcare institutions can purchase licenses to integrate the system into their diagnostic workflow.<br>• Healthcare Partnerships: Collaboration with hospitals, research institutions, and healthcare organizations for deployment.<br>• API-based Services: Providing the detection system as an API that can be integrated into other healthcare applications. |
| 6. | Scalability of the Solution | The proposed system has strong scalability potential.<br>• The system can be expanded to process larger medical datasets for improved |

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| | | accuracy.<br>• It can be deployed on cloud platforms to allow access from multiple healthcare centers.<br>• The solution can be extended to detect other blood-related diseases using similar deep learning techniques.<br>• Mobile or portable diagnostic applications can be developed to allow real-time disease detection in remote areas. |